

STOR 155.010 Lecture 06

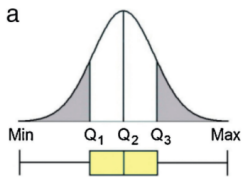
Correlation and Linear Regression I

Aniruddhan (Ani) Ganesaraman

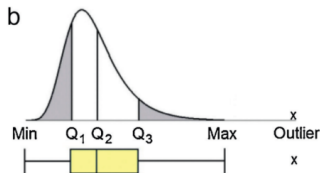
Fall 2026
UNC Chapel Hill

03 September 2026

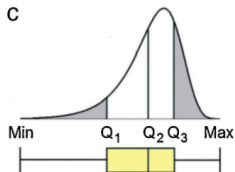
Boxplot and Skewness



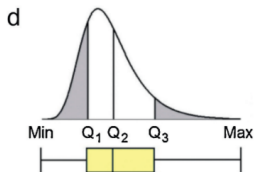
Symmetric



Asymmetric
(positive or right skewed)



Asymmetric
(negative or left skewed)



Asymmetric
(positive or right skewed)

Overview

1 Correlation

2 Linear Regression (Fitting a line)

Introduction

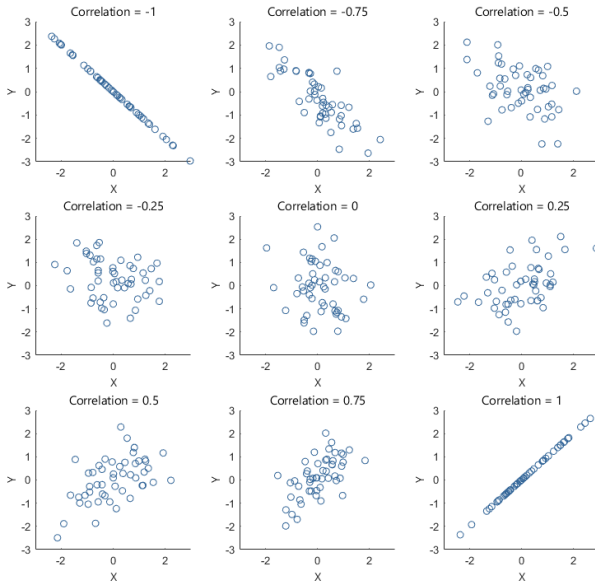
- Measures the strength and direction of a linear relationship between two numerical variables.
- Pearson's correlation coefficient:**

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right),$$

where $s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ and $s_y^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$.

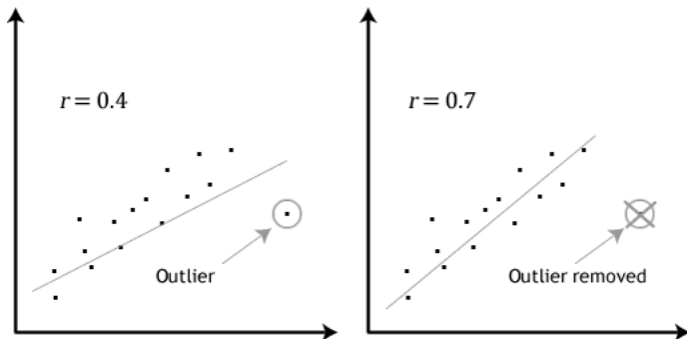
- Always between -1 (strong negative) and 1 (strong positive).
- A value close to 0 implies little to no linear correlation between the variables.

Scatterplots and Correlation



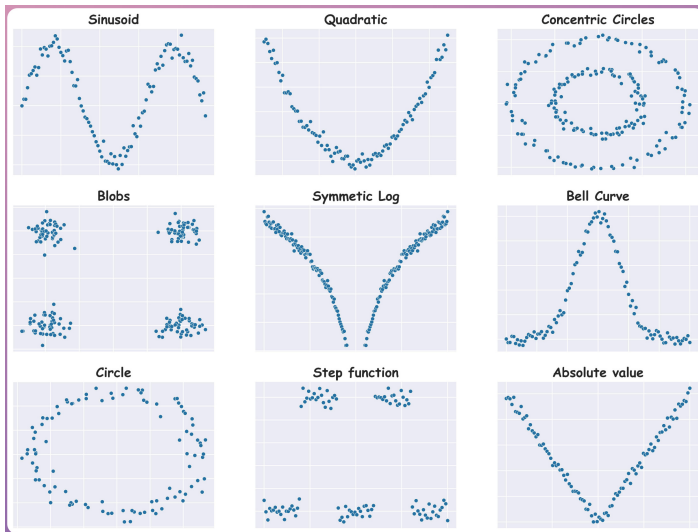
Correlation and Outliers

Correlation is NOT robust to outliers!



Datasets with **zero** correlation

Correlation only measures the linear relationship between two variables.



Limitations

- Correlation measures only linear relationships. Even strong non-linear relationships might have correlation close to zero.
- Correlation is not robust against outliers. A single outlier can significantly change r .
- Correlation does not imply causation. A third factor might influence both variables.
- **Tip.** Always visualize your data with scatterplots to complement the correlation analysis and spot outliers.

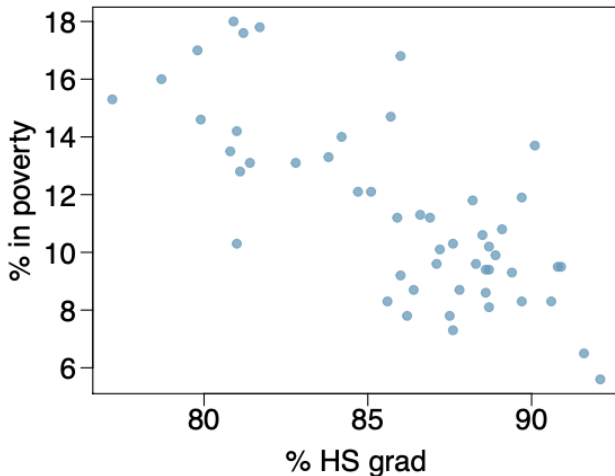
Overview

1 Correlation

2 Linear Regression (Fitting a line)

Poverty vs high school graduation rate

Relationship between HS graduate rate in all 50 US states and DC and the % of residents who live below the poverty line:



Model fitting

- The simplest theoretical model to fit on the data is a **linear model**.
- The best linear model for predicting poverty from high school graduation rate in the US is

$$\widehat{poverty} = 64.78 - 0.62 \cdot HS_{grad}$$

The “hat” is used to signify that this is an estimate.

- The high school graduate rate in Georgia is 85.1%. What poverty level does the model predict for this state?

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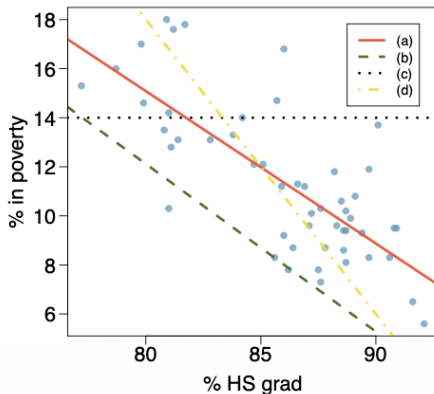
- The high school graduate rate in Georgia is 85.1%. What poverty level does the model predict for this state?

$$64.78 - 0.62 \cdot 85.1 = 12.018$$

- **This ability for prediction showcases the power of having a statistical model!**

Eyeballing the line

Which line best fits the relationship between % in poverty and % HS grad?

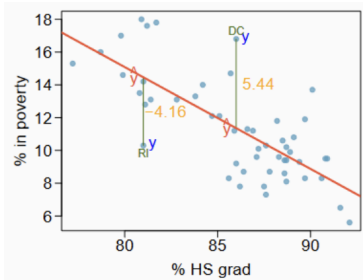
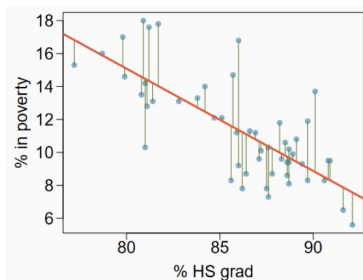


What is it that makes it the best line?

Residuals

Residuals are the leftovers from the fitted model

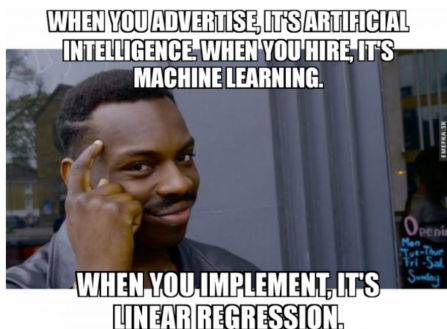
$$\text{Data} \approx \text{Fitted value} + \text{Residual}$$



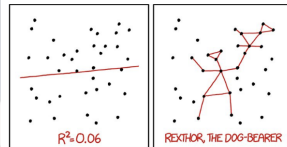
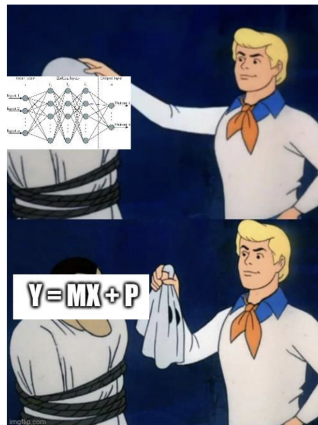
- % living in poverty in DC is 5.44% more than predicted.
- % living in poverty in RI is 4.16% less than predicted.

The “best” line

- Collectively small residuals.
- Minimize $\sum |\text{residuals}|$ or $\sum \text{residuals}^2$.
- The latter leads to **least squares regression**!
- The line that minimizes the sum of squares of residuals is called the *line of best fit*.

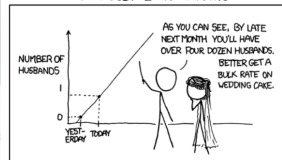


Linear Regression Explainers



I DON'T TRUST LINEAR REGRESSIONS WHEN IT'S HARDER TO GUESS THE DIRECTION OF THE CORRELATION FROM THE SCATTER PLOT THAN TO FIND NEW CONSTELLATIONS ON IT.

MY HOBBY: EXTRAPOLATING



Problem Setup

- Given pairs of data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$.
- We want to construct a model $y_i = f(x_i) + \varepsilon_i$ that connects the variables X and Y .
- We focus on linear functions f :

$$y = f(x) = b_0 + b_1x.$$

- Such a model allows us to predict Y for any value of X .
- X is called the **explanatory** or **independent** or **predictor** variable.
- Y is called the **response** or **outcome** or **dependent** variable.

Finding the line of best fit

- Sum of squares of residuals $SSE = \sum_{i=1}^n (y_i - (b_0 + b_1 x_i))^2$. We want to minimize SSE.
- The best fitting (least squares) line is $\hat{y} = \hat{b}_0 + \hat{b}_1 x$, where

$$\arg \min_{b_0, b_1} \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2.$$

- Technical jargon:
 - $\hat{y}$ is the predicted value of the response variable.
 - $\hat{b}_0$ is the y-intercept.
 - $\hat{b}_1$ is the slope.
 - x is the value of the explanatory variable.

Finding the slope and intercept

- Let $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ and $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$. Also recall $s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ and $s_y^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$.

- Slope:**

$$\hat{b}_1 = r \frac{s_y}{s_x} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

- Intercept:**

$$\hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x}$$

- Once $\hat{b}_0$ and $\hat{b}_1$ are known, any **prediction** for y given a value of x can be made:

$$\hat{y} = \hat{b}_0 + \hat{b}_1 x.$$

References

- Section 8.1 and 8.2 of the textbook
- *Pearson's Correlation, Clearly Explained!!!* by StatQuest
- *Linear Regression, Clearly Explained!!!* by StatQuest